

2024

MUHAMMAD TAHA

22F-3277

BCS-4C

24/03/2024

DATABASE

ASSIGNMENT 3

Q1

1.

```
CREATE TABLE Employee_3277
(
    id      int,
    first_name  varchar(20),
    last_name varchar(20),
    salary  int,
    dob     DATE,
    manager_id  int,
    title   varchar(30),
    joining_date DATE,

    PRIMARY KEY(id)
);
```

Object Type **TABLE** Object **EMPLOYEE_3277**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>EMPLOYEE_3277</u>	<u>ID</u>	NUMBER	22	-	0	1	-	-	-
	<u>FIRST_NAME</u>	VARCHAR2	20	-	-	-	✓	-	-
	<u>LAST_NAME</u>	VARCHAR2	20	-	-	-	✓	-	-
	<u>SALARY</u>	NUMBER	22	-	0	-	✓	-	-
	<u>DOB</u>	DATE	7	-	-	-	✓	-	-
	<u>MANAGER_ID</u>	NUMBER	22	-	0	-	✓	-	-
	<u>TITLE</u>	VARCHAR2	30	-	-	-	✓	-	-
	<u>JOINING_DATE</u>	DATE	7	-	-	-	✓	-	-
1 - 8									

2.

```
INSERT INTO Employee_3277 VALUES (111, 'Muhammad', 'Taha', 900000, TO_DATE('2003-09-23', 'YYYY-MM-DD'), NULL, 'CEO', TO_DATE('2023-01-01', 'YYYY-MM-DD'));
INSERT INTO Employee_3277 VALUES (222, 'Faseeh', 'Shakeel', 800000, TO_DATE('2005-06-10', 'YYYY-MM-DD'), 111, 'Media Team', TO_DATE('2023-02-10', 'YYYY-MM-DD'));
```

ID	FIRST_NAME	LAST_NAME	SALARY	DOB	MANAGER_ID	TITLE	JOINING_DATE
111	Muhammad	Taha	900000	09/23/2003	-	CEO	01/01/2023
222	Faseeh	Shakeel	800000	06/10/2005	111	Media Team	02/10/2023

3.

```
SELECT (id || ',' || first_name || ',' || last_name || ',' || salary || ',' || dob || ',' || manager_id || ',' || title || ',' || joining_date) AS THE_OUTPUT
FROM Employee_3277;
```

THE_OUTPUT
111, Muhammad, Taha, 900000, 2003-09-23, , CEO, 01-01-2023
222, Faseeh, Shakeel, 800000, 2005-06-10, 111, Media Team, 02-10-2023

INSERTING RANDOM DATA FOR QUERING

```
INSERT INTO Employee_3277 VALUES (333, 'Sharjeel', 'Khan', 55000, TO_DATE('1995-09-02', 'YYYY-MM-DD'),
NULL, 'HR Specialist', TO_DATE('2022-09-30', 'YYYY-MM-DD'));
INSERT INTO Employee_3277 VALUES (444, 'Ayesha', 'Akram', 60000, TO_DATE('1990-12-21', 'YYYY-MM-DD'),
222, 'Accountant', TO_DATE('2022-01-10', 'YYYY-MM-DD'));
INSERT INTO Employee_3277 VALUES (555, 'Hurera', 'Khan', 90000, TO_DATE('1999-03-16', 'YYYY-MM-DD'),
333, 'Project Manager', TO_DATE('2019-08-22', 'YYYY-MM-DD'));
INSERT INTO Employee_3277 VALUES (666, 'Qasim', 'Rafay', 75000, TO_DATE('2003-01-05', 'YYYY-MM-DD'),
111, 'Software Developer', TO_DATE('2021-11-01', 'YYYY-MM-DD'));
INSERT INTO Employee_3277 VALUES (777, 'Asad', 'As', 70000, TO_DATE('1986-11-18', 'YYYY-MM-DD'), 111,
'Software Developer', TO_DATE('2020-05-15', 'YYYY-MM-DD'));
INSERT INTO Employee_3277 VALUES (888, 'Ali', 'Haider', 62000, TO_DATE('1992-04-07', 'YYYY-MM-DD'),
NULL, 'Sales Representative', TO_DATE('2023-01-25', 'YYYY-MM-DD'));
INSERT INTO Employee_3277 VALUES (999, 'Noor', 'ul-Ain', 85000, TO_DATE('2004-06-30', 'YYYY-MM-DD'),
222, 'Project Manager', TO_DATE('2018-03-12', 'YYYY-MM-DD'));
```

ID	FIRST_NAME	LAST_NAME	SALARY	DOB	MANAGER_ID	TITLE	JOINING_DATE
111	Muhammad	Taha	900000	09/23/2003	-	CEO	01/01/2023
222	Faseeh	Shakeel	800000	06/10/2005	111	Media Team	02/10/2023
333	Sharjeel	Khan	55000	09/02/1995	-	HR Specialist	09/30/2022
444	Ayesha	Akram	60000	12/21/1990	222	Accountant	01/10/2022
555	Hurera	Khan	90000	03/16/1999	333	Project Manager	08/22/2019
666	Qasim	Rafay	75000	01/05/2003	111	Software Developer	11/01/2021
777	Asad	As	70000	11/18/1986	111	Software Developer	05/15/2020
888	Ali	Haider	62000	04/07/1992	-	Sales Representative	01/25/2023
999	Noor	ul-Ain	85000	06/30/2004	222	Project Manager	03/12/2018

4.

```
SELECT last_name, salary
FROM Employee_3277
WHERE salary NOT BETWEEN 5000 AND 12000;
```

LAST_NAME	SALARY
Taha	900000
Shakeel	800000
Khan	55000
Akram	60000
Khan	90000
Rafay	75000
As	70000
Haider	62000
ul-Ain	85000

9 rows returned in 0.01 seconds

5.

```
SELECT last_name, title, joining_date
FROM Employee_3277
WHERE last_name IN ('Matos', 'Taylor')
ORDER BY joining_date ASC;
```

Results	Explain	Describe	Sql
no data found			

6.

```
SELECT last_name, salary
FROM Employee_3277
WHERE salary BETWEEN 5000 AND 12000
AND id IN (20, 50);
```

There is no any attribute as department_id given in the table. So, if it was, we would replace it with the id in the above line.

Results Expl

no data found

7.

```
SELECT last_name, title
FROM Employee_3277
WHERE manager_id IS NULL;
```

LAST_NAME	TITLE
Taha	CEO
Khan	HR Specialist
Haider	Sales Representative

8. Taking Input from user is not taught.

9.

```
SELECT last_name
FROM Employee_3277
WHERE last_name LIKE '%a%' AND last_name LIKE '%e%';
```

LAST_NAME
Shakeel
Haider

10.

```
SELECT CURRENT_DATE AS "Date" FROM DUAL;
```

Date
03/24/2024

11.

```
SELECT id, last_name, salary, salary*1.155 as New_Salary
FROM Employee_3277;
```

ID	LAST_NAME	SALARY	NEW_SALARY
111	Taha	900000	1039500
222	Shakeel	800000	924000
333	Khan	55000	63525
444	Akram	60000	69300
555	Khan	90000	103950
666	Rafay	75000	86625
777	As	70000	80850
888	Haider	62000	71610
999	ul-Ain	85000	98175

12.

```
SELECT last_name AS "Last Name", LENGTH(last_name) AS Name Length
FROM Employee_3277
WHERE last_name LIKE 'J%' OR last_name LIKE 'A%' OR last_name LIKE 'M%'
ORDER BY last_name;
```

Last Name	NAME_LENGTH
Akram	5
As	2

13.

```
SELECT last_name, LPAD(salary, 15, '$') AS SALARY
FROM Employee_3277;
```

LAST_NAME	SALARY
Taha	\$\$\$\$\$\$\$\$\$900000
Shakeel	\$\$\$\$\$\$\$\$\$800000
Khan	\$\$\$\$\$\$\$\$\$55000
Akram	\$\$\$\$\$\$\$\$\$60000
Khan	\$\$\$\$\$\$\$\$\$90000
Rafay	\$\$\$\$\$\$\$\$\$75000
As	\$\$\$\$\$\$\$\$\$70000
Haider	\$\$\$\$\$\$\$\$\$62000
ul-Ain	\$\$\$\$\$\$\$\$\$85000

14.

```
SELECT Title, COUNT(id) AS No_of_People
FROM Employee_3277
GROUP BY Title;
```

TITLE	NO_OF_PEOPLE
HR Specialist	1
Sales Representative	1
Software Developer	2
Media Team	1
Project Manager	2
Accountant	1
CEO	1

15.

```
SELECT MAX(salary) - MIN(salary) AS DIFFERENCE
FROM Employee_3277;
```

DIFFERENCE
845000

16.

```
SELECT manager_id, MIN(salary) AS Lowest_Salary
FROM Employee_3277
WHERE manager_id IS NOT NULL
GROUP BY manager_id
HAVING MIN(salary) > 6000
ORDER BY Lowest_Salary DESC;
```

MANAGER_ID	LOWEST_SALARY
333	90000
111	70000
222	60000

17.

```
UPDATE Employee_3277
SET salary = 1000
WHERE salary < 900;
```

Results Explain Desci

0 row(s) updated.

0.00 seconds

18.

```
DELETE FROM Employee_3277
WHERE ID = 888;
```

Results Explain [

1 row(s) deleted.

0.00 seconds

19.

```
CREATE TABLE EMPLOYEES2 AS
SELECT ID AS ID, FIRST_NAME AS F_name,
LAST_NAME, SALARY AS Sal
FROM Employee_3277;
```

Results Explain

Table created.

0.01 seconds

Results Explain Describe Saved SQL His

ID	F_NAME	LAST_NAME	SAL
111	Muhammad	Taha	900000
222	Faseeh	Shakeel	800000
333	Sharjeel	Khan	55000
444	Ayesha	Akram	60000
555	Hurera	Khan	90000
666	Qasim	Rafay	75000
777	Asad	As	70000
999	Noor	ul-Ain	85000

20.

```
DROP TABLE Employees2;
```

Results Explain

Table dropped.

0.07 seconds

Q2

1.

```
SELECT Product_Category, AVG(PRODUCT_PRICE) AS Average_Price
FROM Product_Information
GROUP BY Product_Category
HAVING COUNT(*) > 10;
```

Results **Expla**

no data found

2.

```
SELECT EXTRACT(YEAR FROM product_sold_date) AS Year, EXTRACT(MONTH FROM
product_sold_date) AS Month, SUM(product_price) AS Total_Price
FROM Product_Information
GROUP BY EXTRACT(YEAR FROM product_sold_date), EXTRACT(MONTH FROM product_sold_date)
HAVING SUM(product_price) > 10000;
```

Results **Exp**

no data found

3.

```
SELECT product_manufacturer, product_color, MIN(product_price) as Min_Price
FROM Product_Information
GROUP BY product_manufacturer, product_color
HAVING COUNT(*) > 3;
```

Results **Explai**

no data found

4.

```
SELECT sales_person, product_category, SUM(product_price) AS Total_Product_Price
FROM Product_Information
GROUP BY sales_person, product_category;
```

SALES_PERSON	PRODUCT_CATEGORY	TOTAL_PRODUCT_PRICE
Salesperson 3	Category 3	30.99
Salesperson 2	Category 2	46.98
Salesperson 1	Category 1	49.97

5.

```
SELECT sales_person, product_category, product_material, SUM(product_weight) AS
Total_Product_Weight
FROM Product_Information
GROUP BY sales_person, product_category, product_material;
```

SALES_PERSON	PRODUCT_CATEGORY	PRODUCT_MATERIAL	TOTAL_PRODUCT_WEIGHT
Salesperson 1	Category 1	Metal	1.5
Salesperson 2	Category 2	Plastic	4
Salesperson 1	Category 1	Wood	3.5
Salesperson 3	Category 3	Metal	3

6.

```
SELECT EXTRACT(YEAR FROM product_sold_date) AS Year, EXTRACT(MONTH FROM
product_sold_date) AS Month, product_category, SUM(product_price) AS Total_Product_Price_Yearly
FROM Product_Information
GROUP BY EXTRACT(YEAR FROM product_sold_date), EXTRACT(MONTH FROM product_sold_date),
product_category;
```

YEAR	MONTH	PRODUCT_CATEGORY	TOTAL_PRODUCT_PRICE_YEARLY
2022	2	Category 2	20.99
2022	3	Category 1	15.99
2022	4	Category 3	30.99
2022	6	Category 1	22.99
2022	1	Category 1	10.99
2022	5	Category 2	25.99

7.

```
SELECT EXTRACT(YEAR FROM product_sold_date) AS Year, product_category, SUM(product_price)
AS Total_Product_Price_Yearly
FROM Product_Information
GROUP BY EXTRACT(YEAR FROM product_sold_date), product_category;
```

YEAR	PRODUCT_CATEGORY	TOTAL_PRODUCT_PRICE_YEARLY
2022	Category 1	49.97
2022	Category 2	46.98
2022	Category 3	30.99

8.

```
SELECT product_category, sales_person, SUM(product_price) AS Total_Sales, AVG(product_rating)
AS Average_Rating
FROM Product_Information
GROUP BY product_category, sales_person;
```

PRODUCT_CATEGORY	SALES_PERSON	TOTAL_SALES	AVERAGE_RATING
Category 1	Salesperson 1	49.97	4.0666666666666666666666666666667
Category 2	Salesperson 2	46.98	4.3
Category 3	Salesperson 3	30.99	4

9.

```
SELECT product_name, product_sold_date, product_price
FROM Product_Information
WHERE product_sold_date > '01-01-2022'
ORDER BY product_sold_date ASC, product_price DESC;
```

PRODUCT_NAME	PRODUCT_SOLD_DATE	PRODUCT_PRICE
Product A	01/05/2022	10.99
Product B	02/10/2022	20.99
Product C	03/15/2022	15.99
Product D	04/20/2022	30.99
Product E	05/25/2022	25.99
Product F	06/30/2022	22.99

10.

```
SELECT product_name, product_category, sales_person
FROM Product_Information
WHERE Product_Category IN ('Category 1','Category 2','Category 3') AND sales_person IN
('Salesperson 1', 'Salesperson 2')
ORDER BY product_category DESC, product_weight ASC, product_price DESC;
```

PRODUCT_NAME	PRODUCT_CATEGORY	SALES_PERSON
Product E	Category 2	Salesperson 2
Product B	Category 2	Salesperson 2
Product C	Category 1	Salesperson 1
Product A	Category 1	Salesperson 1
Product F	Category 1	Salesperson 1

11.

```
SELECT product_category, SUM(product_price) AS Total_Sales_Amount
FROM Product_Information
GROUP BY product_category
HAVING SUM(product_price) > 100;
```

Results Expla

no data found

12.

```
SELECT product_manufacturer, AVG(product_rating) AS Average_Rating
FROM Product_Information
GROUP BY product_manufacturer
HAVING AVG(product_rating) > 4;
```

PRODUCT_MANUFACTURER	AVERAGE_RATING
Manufacturer 1	4.1
Manufacturer 2	4.3

13.

```
SELECT sales_person, COUNT(product_id) as Products_Sold
FROM Product_Information
GROUP BY sales_person
HAVING COUNT(product_id) > 2;
```

SALES_PERSON	PRODUCTS_SOLD
Salesperson 1	3

14.

```
SELECT product_category, AVG(product_weight) AS Average_Weight
FROM Product_Information
GROUP BY product_category
HAVING AVG(product_weight) < 2;
```

PRODUCT_CATEGORY	AVERAGE_WEIGHT
Category 1	1.66666666666666666666666666666667

15.

```
SELECT Product_Name, Product_Rating, Sales_Person  
FROM Product_Information  
WHERE Product_Rating > 1 AND Sales_Person = 'Salesperson 1';
```

PRODUCT_NAME	PRODUCT_RATING	SALES_PERSON
Product A	4.2	Salesperson 1
Product C	4.5	Salesperson 1
Product F	3.5	Salesperson 1